PROGRAM:

AIES-Exp_3.py - C:/Users/ABBIRAMY/OneDrive/Documents/AIES_LAB/AIES-Exp_3.py (3.14.2)

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```
import heapq
graph = {
    'S': {'A': 1, 'B': 4},
    'A': {'B': 2, 'C': 5, 'D': 12},
    'B': {'C': 2},
    'C': {'G': 3},
    'D': {'G': 3},
    'G': {}
}
h = {'S': 7, 'A': 6, 'B': 4, 'C': 2, 'D': 3, 'G': 0}
def a_star(graph, h, start, goal):
    pq = [(h[start], 0, start, [start])]   # (f, g, node, path)
    best_g = {}
    while pq:
        f, g, node, path = heapq.heappop(pq)
        if node == goal:
            return path, g
        if node in best_g and best_g[node] <= g:
            continue
        best_g[node] = g
        for nxt, w in graph[node].items():
            g2 = g + w
            heapq.heappush(pq, (g2 + h[nxt], g2, nxt, path + [nxt]))
    return None, float('inf')
path, cost = a_star(graph, h, 'S', 'G')
print("Optimal path:", path)
print("Total cost  :", cost)
```

OUTPUT:

IDLE Shell 3.14.2

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```
Python 3.14.2 (tags/v3.14.2:df79316, Dec  5 2025, 17:18:21) [MSC v.1944 64 bit (
AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Users/ABBIRAMY/OneDrive/Documents/AIES_LAB/AIES-Exp_3.py =====
Optimal path: ['S', 'A', 'B', 'C', 'G']
Total cost  : 8
>>>
```